

Adora — Plan A: Synthesising Spectra from 3D Cubes

Extension plan · Ivan Milic · 2026-08-28 · 2/3

Goal. Feed a 3D cube of physical parameters $(n_x, n_y, n_z) — T, n_e, n_H, v_z, v_{\text{turb}}, B, \gamma_B, \chi_B$ — and produce a Stokes cube $(n_x, n_y, 4, n_\lambda)$. **Strategy: start 1.5D (column-by-column vertical rays)**, since the RT is already 1D along z — this is the cheap, high-value first step; true 3D (lateral) transfer is a separate, larger effort deferred to v2.

- **Ingestion.** Loader for snapshots (MURaM/Bifrost) $\rightarrow$ the 8 atmospheric arrays per column, in the code's units and top-down ordering; per-column geometric dz (or a common grid). Optionally resample onto a $\log \tau$ grid for accuracy near line-forming heights.
- **Vectorisation.** One column is exactly today's `lte_polarised_rt` (already vmapped over λ). Wrap in nested `jax.vmap` over $(x, y) \Rightarrow$ batched Stokes cube; `jit` the per-column $\times$ per- λ kernel. Tile columns to bound memory; `shard_map/pmap` across GPUs. Precompute `AtomicData` once; stream output to `zarr/npz`.
- **Physics upgrades needed for real cubes.** (i) EOS to get n_e, n_{HI} consistently when cubes give only (T, ρ) or (T, P) ; (ii) more elements' partition functions; (iii) optional inclined rays $\mu < 1$: project v_z and $\mathbf{B}$ onto the LOS and integrate along the slanted path (note: the polarised opacity currently assumes $\mu_z = 1$ and must be generalised).
- **Deliverable & validation.** A [synth3d.py](#) (cube + line list + λ -grid $\rightarrow$ Stokes cube, optional response cube via `vmap(jacrev)`); spot-check columns against RH/Lightweaver (continuum, line depths, V amplitude).